

FIN 294 Problem Set 1

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MEMO

Questions & Solutions

Problem 1

You have the option of receiving a one-time lump sum bonus of \$1 million in 10 years. If the relevant opportunity cost of capital (discount rate) is 8%, what is that bonus worth today?

In order to know what today's value of the \$1 million lump sum bonus is, you need to take the future value and discount it back ten years by using the 8% discount rate and $n=10$. Take the future value (\$1,000,000) and divide it by the sum of 1 and the discount rate (8%) —adding 1 to signify a one year growth — to the power of 10. Therefore, the present value of the \$1,000,000 in 10 years is **\$463,193.49**.

Problem 2

Suppose now that you have the option of receiving the one-time lump sum bonus of \$1 million in 10 years or \$350,000 today. What is the breakeven required return you'd need on the payment today to be indifferent regarding the two choices? Assuming both payments are relatively secure (there is no default risk), which one would you choose? Why?

In order to make the smartest decision here, you need to bring the two values to the same time period to make a fair comparison. At first glance, the present value (answer to problem 1) exceeds \$350,000 by about \$115,000. Therefore, I would choose to receive \$1,000,000 in 10 years. The second part of the question requires you to algebraically rearrange the formula used above to solve for the discount rate to find the break even required return. It's asking what discount rate is needed for the \$350,000 to be exactly \$1,000,000 in 10 years. The future value (\$1,000,000) divided by the present value (\$350,000) all to the power of $1/10$ minus 1 brings us to a breakeven required return rate of **11.07%**. This rate is much higher than the 8% from problem 1, so the return is much higher as well.

Problem 3

A lottery currently has an advertised jackpot of \$100 million. However, that is paid out as a series of 30 annual payments of \$3,333,333. Alternatively, you could elect to receive \$47 million today instead. Assuming you could invest at 5%, which alternative should you choose?

Similarly to the other problems in the set, you need to bring the \$100,000,000 to the present by discounting in order to make a valid assessment of your options. Since it is a series of annual payments in a fixed period of time (30 years), we use the formula for the present value of an annuity. We take \$3,333,333 times 1 minus the quotient of 1 over 1.05% to the power of 30 (for the 30 years), all divided by 5%. The resulting value is **\$51,241,498.30**. You could either take \$47 million now, or you can play the long game and make an additional ~\$4 million through a series of 30 annual payments compounded over time. I would choose the latter.

Problem 4

Considering the lottery in Problem III, what would be the breakeven rate of return to be indifferent regarding the two options?

For Problem 4 we were given an original rate of 5% we need to find the breakeven rate of return to be indifferent.

Use start with the same formula we used for problem 3 but this time we have the PV(present value) of \$47,000,000 dollars, CF1(cash flow 1) of \$3,333,333 dollars(1 of 30 payments), g (annuity growth rate) of 0 as (Stays \$3,333,333 for each of 30 payments), n(number of payments) being 30. We're only missing the r discount rate. We used the Goal Seeker function in Excel to select the discount rate using the same formula as above with \$47,000,000 instead of the \$51,241,498. This outputted a value of **5.78%** for the discount rate.

Problem 5

Assume that you just received a cash flow of \$1,000 (CF_0). For the next 5 years (CF_1 through CF_5), you expect the growth rate to be 10%. After that period, you expect the growth rate to slow to 4%, but the cash flows will continue indefinitely (perpetually). What is the present value of these cash flows? HINT: Do not include CF_0 in the estimate because you have already received it. Assume a discount rate of 12%.

To find the present value of all of the cash values in this problem, you need to calculate the annuity separate from the perpetuity, then discount them all to the present and add

up the values. From CF1 to CF5, you find the present value of the annuity by taking the cash flow at time period=0 times the sum of 1 and the growth rate divided by the discount rate (12%) minus the growth rate (10%). Then, multiply that by 1 minus 1.1% (1+growth rate) divided by 1.12% (1+discount rate) to the power of 5 for the number of periods. Therefore, the present value of the annuity is \$4,738.44. In order to find the present value of the perpetuity, you need to first calculate the cash flow at time period=6 (when the growth slows down), find the compounded value of the perpetuity, then you can discount it back to the present. Simply put, the cash flow at time period=6 can be calculated by taking the cash flow at time period=5 (\$1,610.51) and multiplying it by 1.04% (1+new growth rate). The cash flow at time period=6 is \$1,674.93. Apply the growing perpetuity formula → CF6 divided by 12% minus 4% to get \$20,936.63. Discount this projected value of the perpetuity → \$20,936.63 divided by 1.12 to the power of 5, resulting in \$11,880.01. Finally, the present value of each of these cash flows is just the sum of the PV annuity and PV perpetuity: **\$16,618.54.**

Problem 6

As a demonstration of this, show how this program/dashboard solves Problem 5.

Does it agree with your manual solution? Explain any differences.

Dashboard photos for solving problem 5:

Valuation Calculator

Financial Analysis, Assignment 1: Problem 6

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July 19th, 2026

Select a Calculation:

☒ Present Value

☐ Future Value

Number of discrete forecast periods

5

Number of growth phases (including 'none')

1

Discount rate (%)

12.00

Initial cash flow (CF0)

1000.00

Cash flow in period 1

1100.00

Cash flow in period 2

1210.00

Cash flow in period 3

1331.00

Cash flow in period 4

1464.10

Cash flow in period 5

1610.51

Phase 1

Enter the growth rate for phase 1 as a percentage

4.00

Annuity or Perpetuity?

☐ Annuity

☒ Perpetuity

Perpetuities have no defined length

Phase Inputs

Phase	Growth Rate	Length	Terminal Case
1	4.00	0.00	Perpetuity

Valuation Breakdown

Component	Value
1. PV Discrete Forecast	4784.4355
2. Present Value Phase 1 Perpetuity	11885.0041

Result

Total Present Value

\$16,618.44

Our dashboard does agree with our manual solutions; we both reached a total present value of \$16, 618.44

What benefits and limitations does this program have vis-à-vis your manual work?

Benefits:

Being able to simply input the numbers and receive an output speeds up the computation time and prevents calculation errors since it loads the correct inputs automatically. This is especially helpful when working with multiple phases that differ slightly since it automatically iterates through them instead of having to manually calculate it for each phase.

If you are working in the same session, then you can easily change single values and reuse the other inputs without having to redo the whole calculation.

We also structured it so that you can see a breakdown of each input at each step, which makes it easier to understand and review the output.

Limitations:

You have to manually enter each of the period cash flows by hand, which opens itself to errors either from misunderstanding which inputs correspond to which cash flows or from manual calculation errors.

Another limitation is that, during development, everytime the output didn't match the manual calculation we had to go back through all the code to find the bug and then fix that and retest it.

You still need to be able to verify the output number (especially when testing) since the program outputs a number without any notice of error.

References and LLM Prompts:

References:

"Input Widgets." *Streamlit Documentation*, Streamlit, <https://docs.streamlit.io/develop/api-reference/widgets>. Accessed 19 July 2026

"Data Elements." *Streamlit Documentation*, Streamlit, <https://docs.streamlit.io/develop/api-reference/data>. Accessed 19 July 2026.

Claude Prompts used for dashboard:

- Write some test inputs and provide the correct value for me to compare with my dashboard output.
- Using to debug -> what part of this code block is preventing my dashboard from calculating anything?
- How do I make my label dynamic so that the user doesn't have to know that the initial cash flow doubles as the anchor?
- What part of my code is producing the incorrect calculation for a growing PVA. My manual calculations include: $2000 \times 0.20754 = \$415.08$, My dashboard outputs \$435.81
- How do I make the length of phase dynamic based on if the user selects annuity or perpetuity?